

EMILY SIMON

William Eckhardt Research Center
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EDUCATION

University of Chicago, Chicago, IL

Ph.D. in Astrophysics, September 2021- expected 2026

NSF GRFP Fellow

Elaine K. Bernstein Women in Science Award Recipient

Columbia University, New York, NY

B.A. in Astrophysics, Concentration in Math, September 2018-May 2021

De Lara Scholarship

Phi Theta Kappa Scholarship

Dean's List Fall 2018

Borough of Manhattan Community College, New York, NY

Major in Physics, September 2016-May 2018

Dean's List, Recognized as BMCC Achiever for 4.0 GPA

RESEARCH

Pre-candidacy Research, University of Chicago, September 2021-present

Principle Investigator: Damiano Caprioli, University of Chicago

Collaborators: Brian Reville (Max Planck Institute for Nuclear Physics), Colby Haggerty (University of Hawaii Institute for Astronomy)

Using hybrid particle-in-cell simulations to study the maximum cosmic ray energy achievable via diffusive shock acceleration at supernova shocks.

Senior Thesis, Columbia University, September 2020-May 2021

Principle Investigators: Lorenzo Sironi and Luca Comisso, Columbia University

Investigating turbulence as a possible acceleration mechanism for ultra-high energy cosmic rays.

CERN REU, (Remote due to COVID-19), May 2020-August2020

Principle Investigator: John Parsons, ATLAS Experiment, Columbia University

Search for beyond the Standard Model physics through an analysis of non-pointing and delayed photons produced in secondary vertices of Higgs decays.

ATLAS Internal Note: "Search for non-pointing and delayed photons with ATLAS in pp collisions at $\sqrt{s} = 13$ TeV"

Columbia University, New York, NY June 2019-September 2019

Principle Investigators: Lorenzo Sironi and Luca Comisso, Columbia University

Magnetic reconnection in turbulent astrophysical plasmas as a driver of nonthermal particle emission.

American Museum of Natural History, New York, NY June 2018-January 2019

Principle Investigator: Quinn Minor, CUNY/BMCC/AMNH

Strong gravitational lensing as a probe of dark matter density in simulated lens systems.

COMPUTER SKILLS

Proficient in: Python, MATLAB, Fortran; **Experience with:** C++, ROOT

Additional skills: Machine Learning basics

Comfortable with Mac, Windows, and Microsoft Office Suite

RELEVANT COURSES TAKEN

Math	Physics	Astronomy
• Multi-Variable Calculus • Linear Algebra • Ordinary Differential Equations • Partial Differential Equations • Mathematical Methods of Physics	• Mechanics • Electricity and Magnetism • Thermal and Statistical Physics • Quantum Mechanics • Particle Physics • General Relativity	• Stars • Interstellar Medium • Galaxies • High Energy Astrophysics • Cosmology I, II, Particle Astrophysics & Cosmology • Detection of Radiation

TEACHING EXPERIENCE

Physics of Stars: An Introduction, Instructor, July 2022

Supervisor: Rich Kron, University of Chicago

Designed and taught an introductory Astronomy class for a three-week immersive high school program at the University of Chicago involving lectures, labs, and computer programming.

Columbia Tutoring and Learning Center (TLC), Lead Tutor, February 2019-2021

Tutoring Geometry at high schools in Harlem, New York

Attending weekly lectures about pedagogy and tutoring practices to promote student agency and engagement

Class Grader: Stars Galaxies and Cosmology, September 2020 – present
Reviewing homework for class with over 70 students, giving feedback, and assigning grades

WORKSHOPS, SHOOLS, & FELLOWSHIPS

Foundations of Cosmic Ray Astrophysics, International School of Physics “Enrico Fermi”, Varenna, Italy, June 23-29, 2022

CERN REU Particle Physics Course, University of Michigan, Professor Jianming Qian, June 2020-August 2020

ROOT and MCMC Summer Workshop, University of Michigan, Junjie Zhu & Alexis Mulski, June 2020-August 2020

Data-Driven Models, Computational Workshop, Lauren Anderson, Flatiron Institute, May 2019

CUWiP Princeton Plasma Physics Workshop, Princeton, New Jersey, January 2019

AstroCom NYC Fellowship, The American Museum of Natural History, January- August 2018

PAPERS & PRESENTATIONS

Quinn Minor, Manoj Kaplinghat, Tony H Chan, **Emily Simon**, Inferring the concentration of dark matter subhaloes perturbing strongly lensed images, MNRAS, July 2021
<https://doi.org/10.1093/mnras/stab2209>

Columbia Nevis Labs, Final Presentation, online
“Search for SUSY with delayed and non-pointing photons,” August 2020

CERN REU, Final Presentation, online
“Search for SUSY with delayed and non-pointing photons,” August 2020

AstroFest, Columbia University, New York, NY
“Magnetic Reconnection as a Potential Driver of Non-Thermal Particle Emission in Astrophysical Plasmas,” September 2019
Awarded 1st Prize Poster Presentation

American Astronomical Society 233rd Meeting, Seattle, WA
“Perturbations of Gravitational Lenses Due to Low-Mass Dark Matter Halos,” January 2019

AstroCom Final Presentation, American Museum of Natural History, New York, NY
“The Effect of Redshift on Inferred Mass of Dark Matter Halos,” July 2018